Rear Wing Stanchion Material Type and Mounting Position on Rear Automotive Spoiler

Jeffrey Bratman [1]

[1] *Mechanical Aerospace Engineering, University of California San Diego*
Engineers Ln, San Diego, CA, United States of America
[1] jbratman@ucsd.edu

Abstract— **The purpose of this study is to investigate the structure of the rear wing stanchion of an automotive rear spoiler. The rear wing stanchion, being the part that holds the wing element to the vehicle is subjected to multiple different loading conditions. This study focuses on the loading that is caused by the downforce that is produced by the wing element on to the stanchion. The investigation looks at the amount of material and type needed to withstand these aero loads vs the type of stanchion design used. Specifically looking at the two main designs of stanchions, one being the standard under wing element, and the other being called the swan neck, which mounts to the upper side of the wing element.**

I. Introduction

Aerodynamic properties of vehicles are increasingly important when designing a vehicle. This is also true in the automotive racing area, where vehicle aerodynamics can be the difference between a win or loss. As a vehicle is driven at high speeds, the air flowing over the rear of the vehicle begins to separate from the vehicle surface. This will begin to cause the rear of the vehicle to rise, reducing the stability of the vehicle and handling potential [1].

With the installation of a rear spoiler on the vehicle, a wing element is installed into the air stream above the separation boundary of the air flow and vehicle body. This inverted wing generates a downward force that pushes the rear of the vehicle into the ground and increases stability and handling potential. This downward force is then transferred into the mounting stanchions. The force transferred into the stanchions is generated by the wing element's ability to generate lift. In the case of an automotive spoiler the lift is towards the ground. The strength of the stanchions is therefore a critical element that needs to be considered when designing a rear spoiler. Failure in the stanchions can lead to unforeseen aerodynamic effects on the vehicle.

The increasing ability of vehicles to achieve higher speeds than previously designed vehicles require more careful consideration of the aerodynamic loads of the vehicle. The first racing car that made use of this rear wing idea was the Chaparral 2C in 1965. In the 1970s front and rear spoilers made their way to production sport cars [2]. As the development of vehicles has advanced so too has the development of the aerodynamic systems that help to maintain stability and handling dynamics of the vehicle.

The development of the rear spoiler stanchion is a result of the increased aerodynamic ability of the rear wing element and progressive improvements to the design. The standard way of mounting the rear wing element to a vehicle is by using vertical stanchions that attach to the underside of the wing element. Due to the design of the wing element however, this places the stanchion on the wing in an area that would be generating low pressure. This could decrease the wing performance by introducing separation of the air from the wing element. A new stanchion design was developed which moves the mount to the top section of the wing and into the high-pressure area, which gives the wing element more surface area on the underside to generate a pressure differential, allowing the wing to increase performance with an overall smaller area.[3]

These two different designs in rear wing stanchion design also come with their own unique design considerations when it comes to the material choice. This study is an investigation of potential material choices, and the differences needed for each type of mount. The wing element used in this study is an unspecific wing design that is of generic wing element shape and is used to determine a constant aero load for the stanchions to be put under for testing.

II. Materials and Method

A. Materials

In this study there are three different material choices that are going to be tested for each type of stanchion. To do this the material properties will be required, Young's modulus, density, tensile strength, and yield strength. For this study the materials in question are Alloy Steel, 6061 Aluminum Alloy, and a Carbon Fiber Epoxy Composite. The mechanical properties of these materials used in the simulation are listed in Table 1. The properties details are acquired from the SolidWorks Library and MatWeb [4].

TABLE I
Mechanical Material Properties

Material	Density (kg/mm³)	Shear modulus (GPa)	Tensile Strength (MPa)	Yield Strength (MPa)
Alloy Steel	7700	7.9	723.8256	62.0422
Aluminum Alloy (6061)	2700	69	124.084	55.1485
Carbon Fiber Epoxy Composite [4]	1150	1.9	4.62	1.0

B. *Modelling and Simulation*

Modeling of the rear spoiler stanchions was done in SolidWorks. To begin with a generic spoiler was acquired from the GrabCAD Community Library [5]. This spoiler was then modified to have an overall width of 1700mm and an increase of blade width to 304mm. The wing was modeled to be at an Angle of Attack of 2°relative to the ground plane. This was done to mimic a real-world rear spoiler section and allow for flow testing. The purpose of flow testing the wing was to get an aerodynamic load that would be transferred into the wing stanchions.

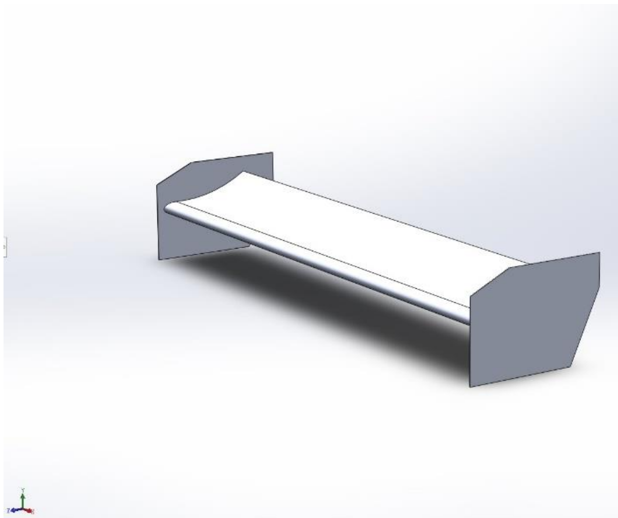

Fig. 1. Model of wing element used in flow testing.

With the wing element fully modeled, the stanchions could be modeled using the wing as a reference to allow them to be contoured to the wing blade profile. With the flow analysis of the wing complete and the standard style and swan neck style stanchions modeled, the static analysis of the stanchions could be completed. The thickness of the main body of both standard and swan neck stanchions are 0.25 in wide with a mounting plate also being the same thickness and having an overall length of 174.15mm. This minimizes the difference between the two different styles of mounts and allows the testing conditions to be equal.

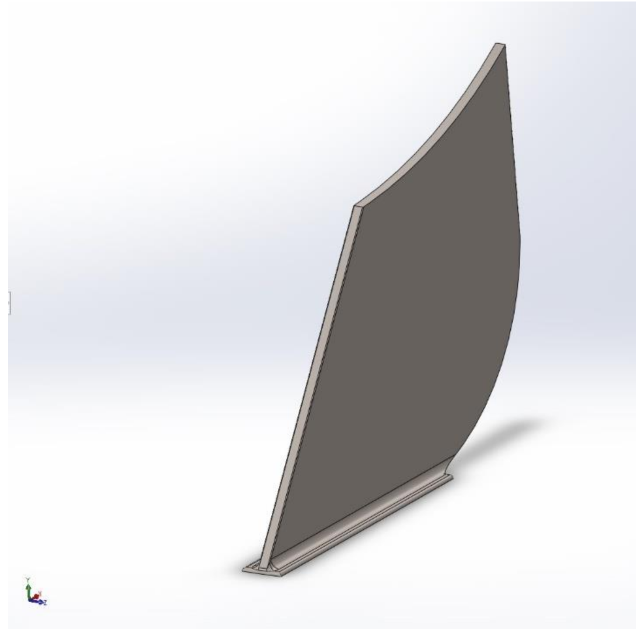

Fig. 2. Under wing Stanchion Model

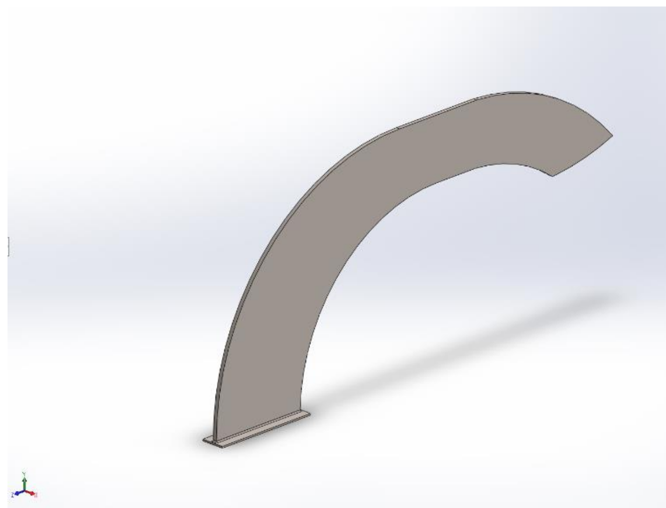

Fig. 3. Swan Neck Stanchion Model

III. Results and Discussions

The Finite Element models of the two rear wing stanchions are subjected to a downward force of 800 N, on the mounting faces where the wing would be mounted to. The studies done on each stanchion are stress, displacement, and factor of safety. With a factor of safety being 3, due to the increase in loading as the vehicle increases speed. This will allow the wing to potentially generate a maximum of 2400 N or ~ 540 lbf of downforce before failure.

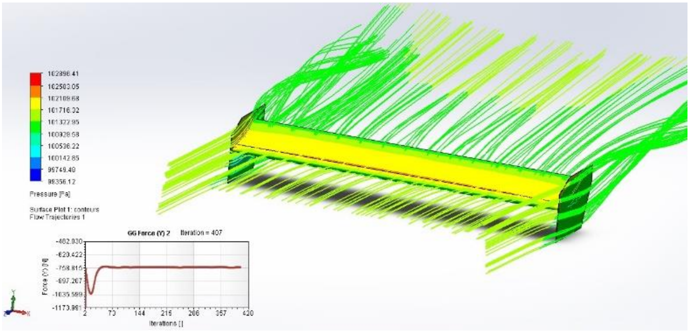

Fig. 4. Flow analysis of generic wing

From the flow analysis done on the standard wing, the maximum force in the downward direction with a flow of 45 m/s (~100 mph) ended up being about 750 N which is less than the 800 N the stanchions were tested at. The following figures 5-22 are the results of the FE of both stanchions in stress, deformation and factor of safety.

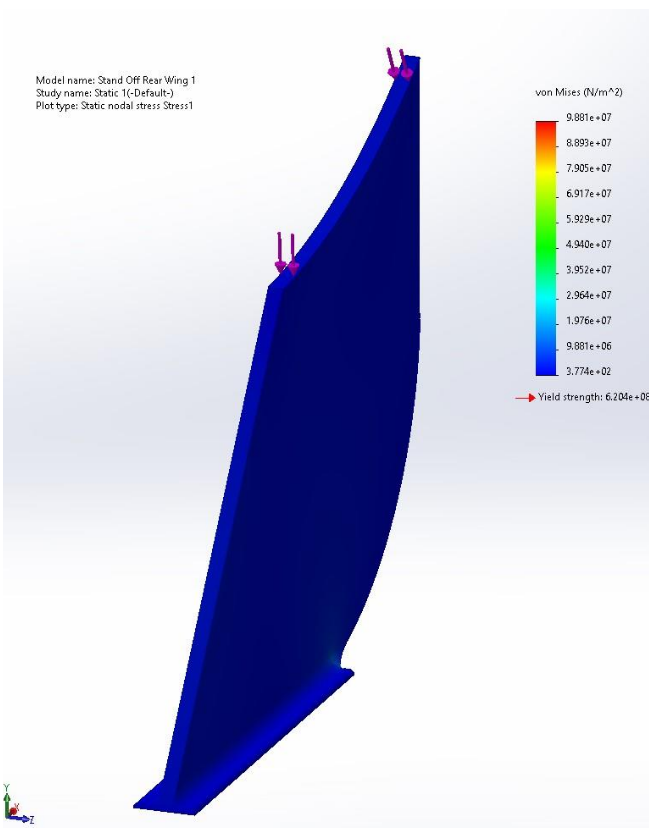

Fig. 5. Stress Plot Alloy Steel

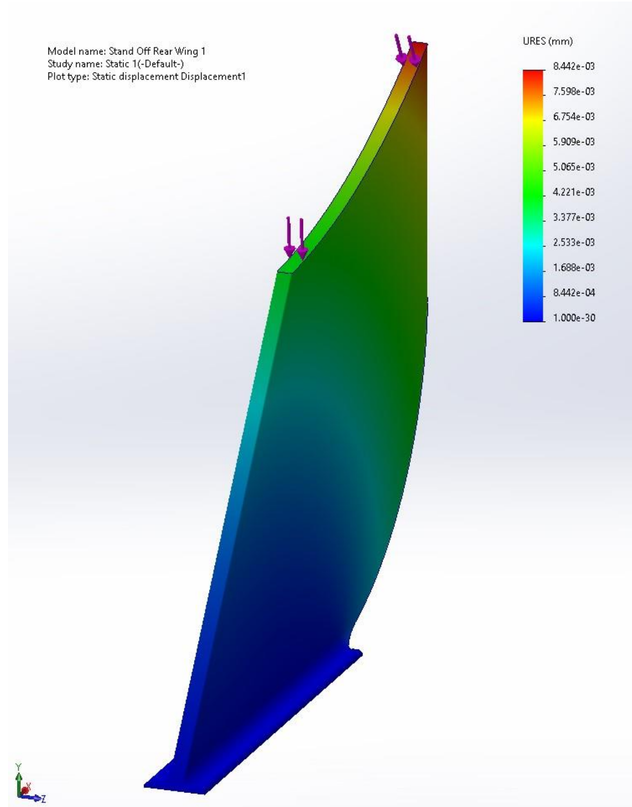

Fig. 6. Displacement Plot Alloy Steel

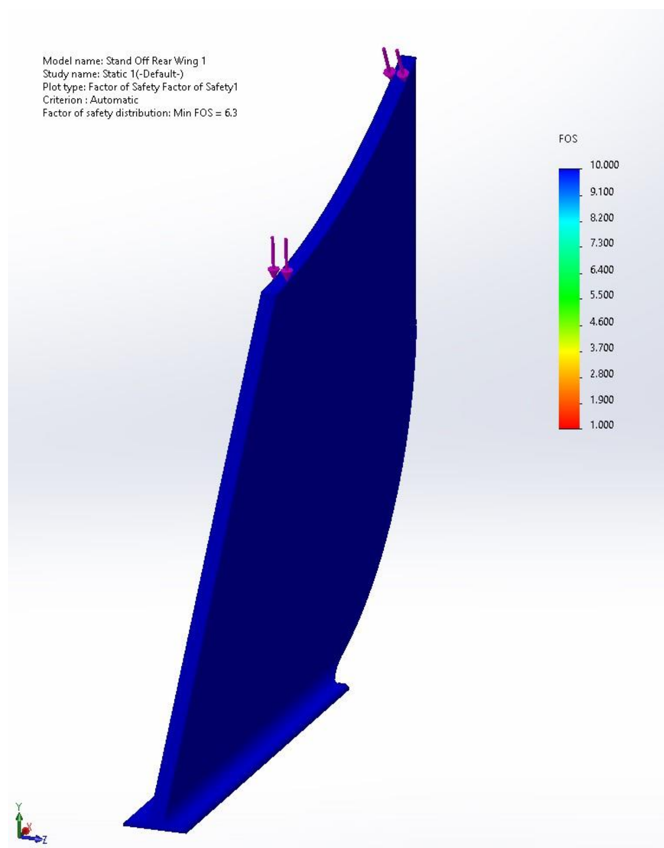

Fig. 7. Factor of Safety Plot Alloy Steel

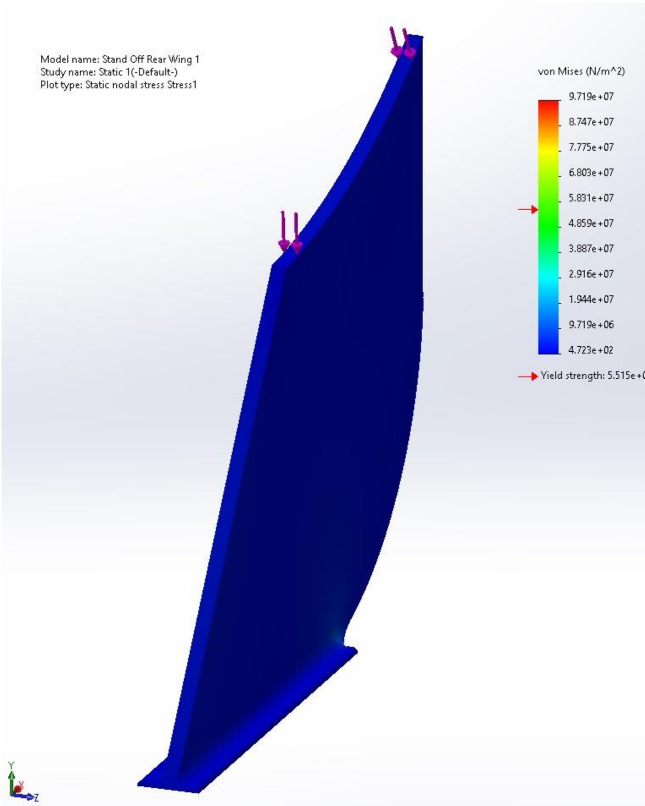

Fig. 8. Stress Plot Aluminum Alloy 6061

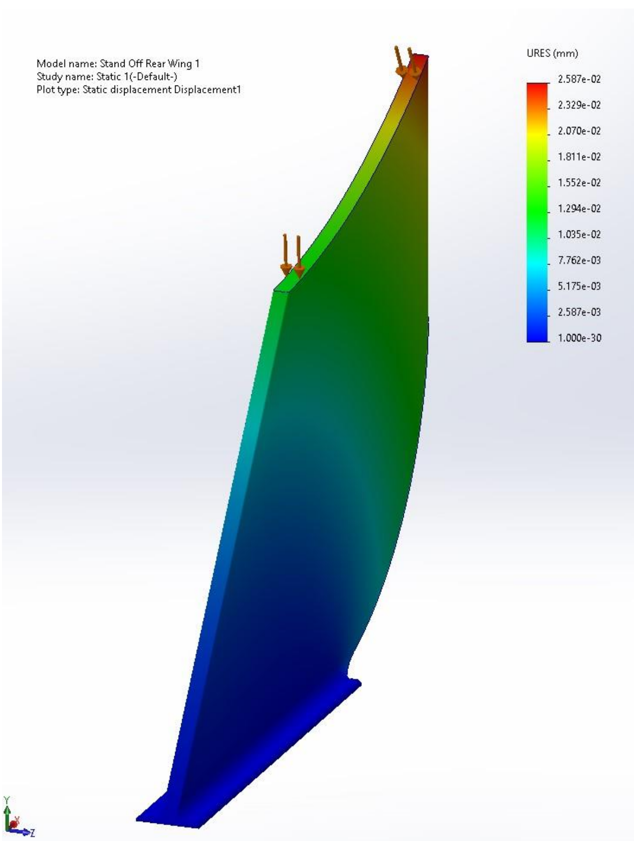

Fig. 9. Displacement Plot Aluminum Alloy 6061

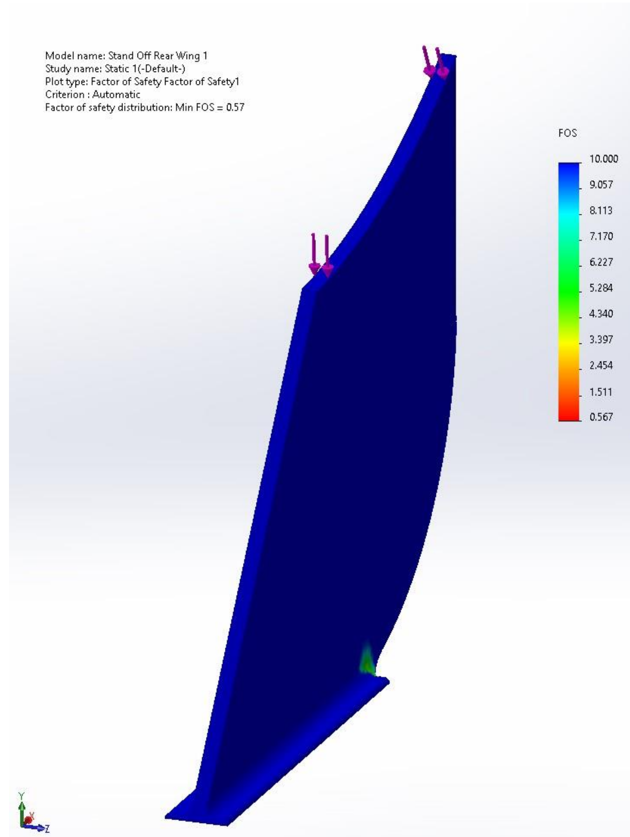

Fig. 10. Factor of Safety Plot Aluminum Alloy 6061

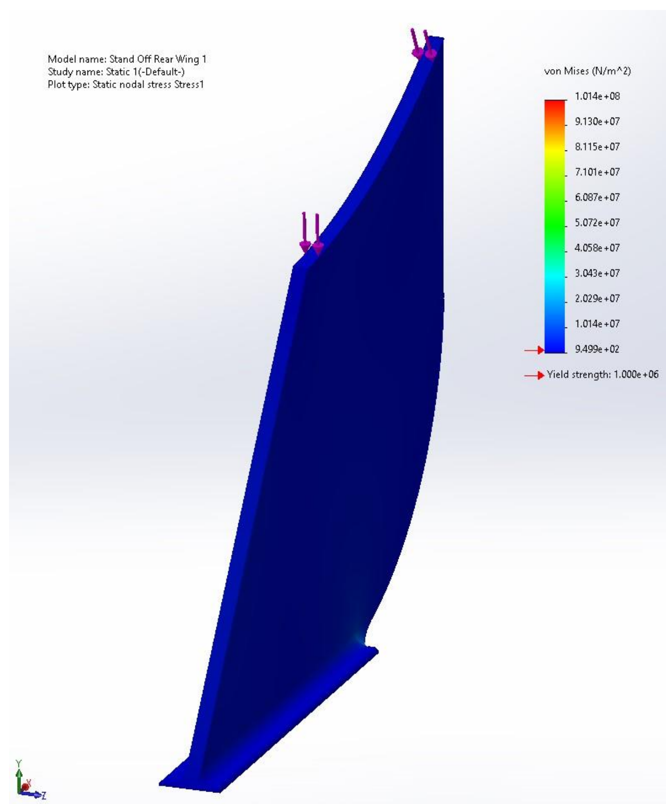

Fig. 11. Stress Plot Carbon Fiber Epoxy

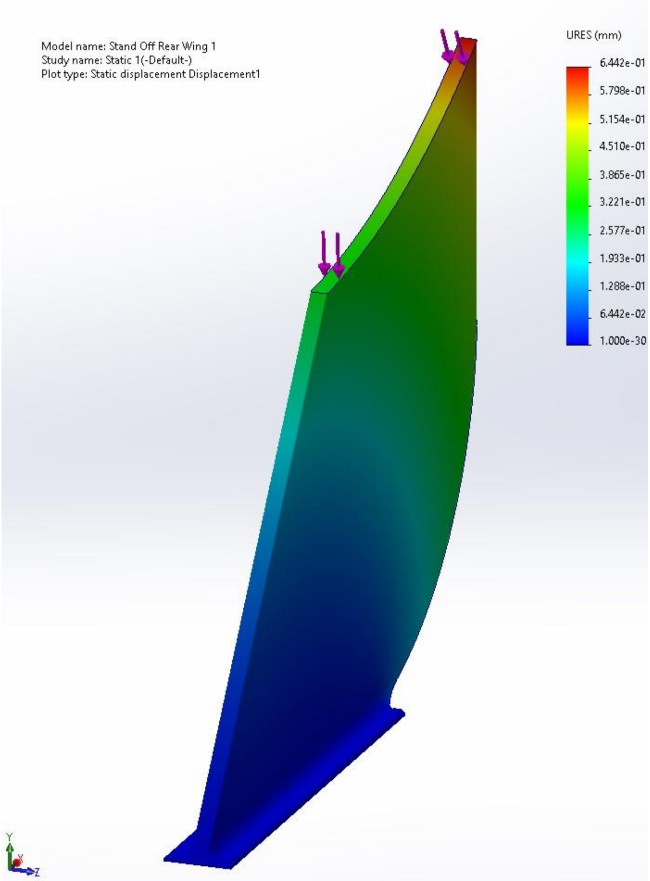

Fig. 12. Displacement Plot Carbon Fiber Epoxy

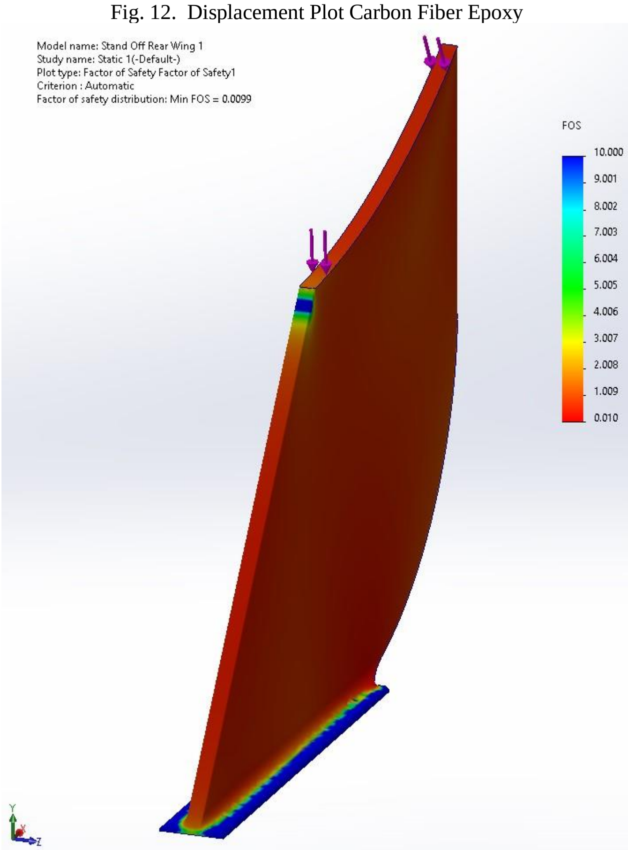

Fig. 13. Factory of Safety Plot Carbon Fiber Epoxy

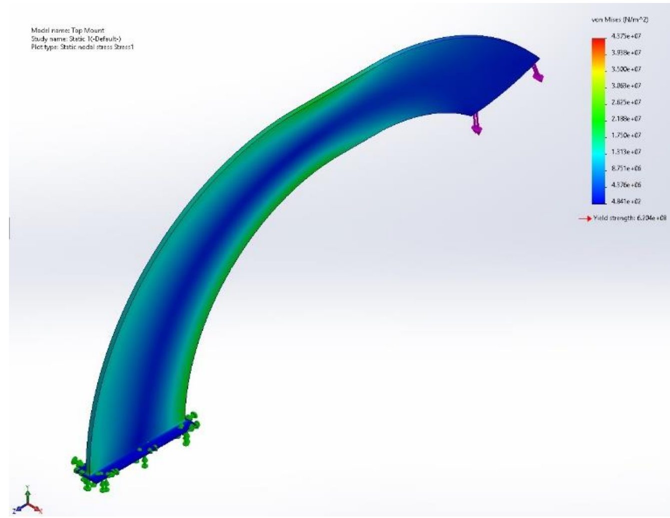

Fig. 14. Stress Plot Alloy Steel

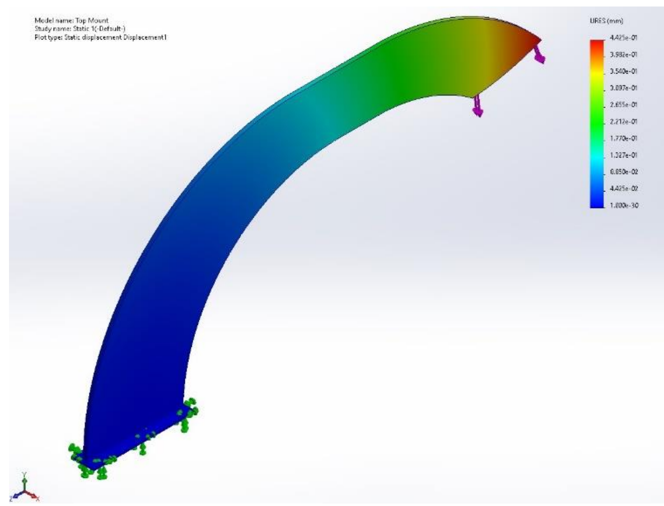

Fig. 15. Displacement Plot Alloy Steel

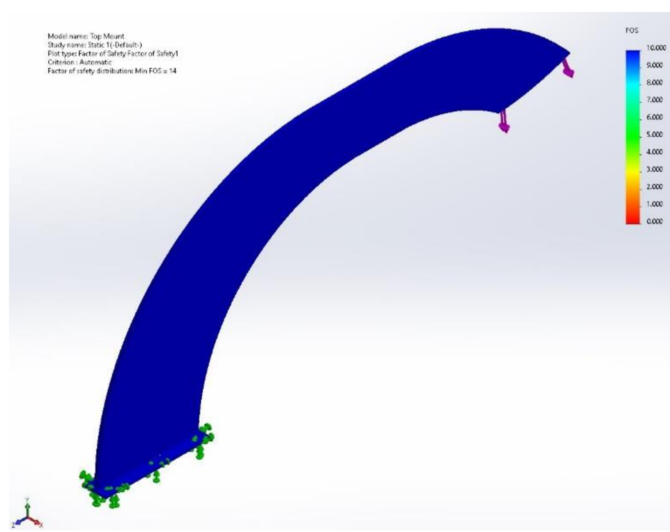

Fig. 16. Factor of Safety Plot Alloy Steel

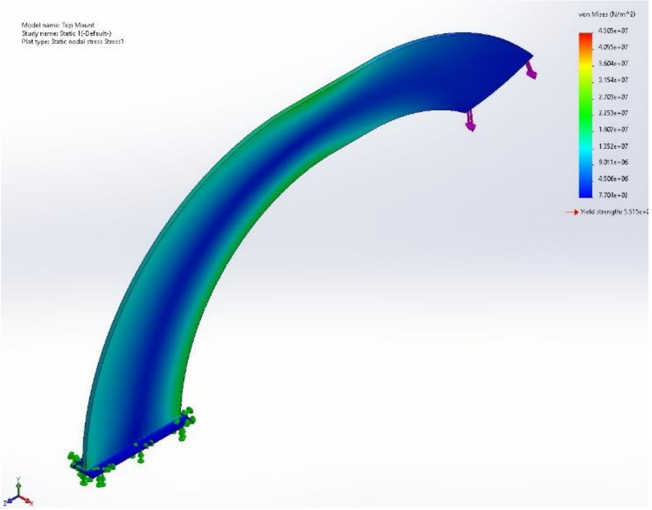

Fig. 17. Stress Plot Aluminum Alloy 6061

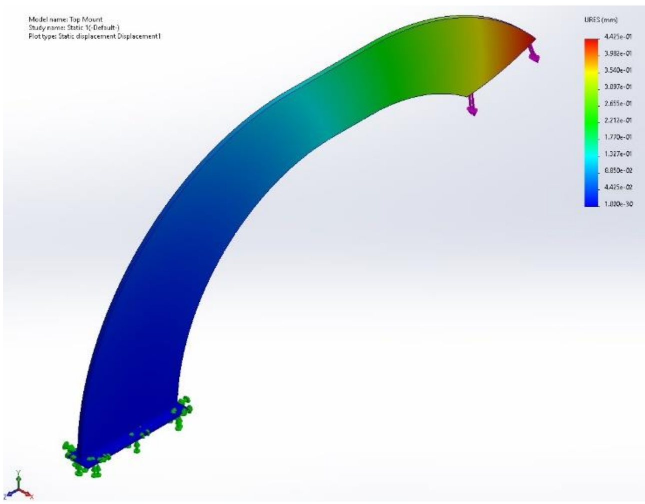

Fig. 18. Displacement Plot Aluminum Alloy 6061

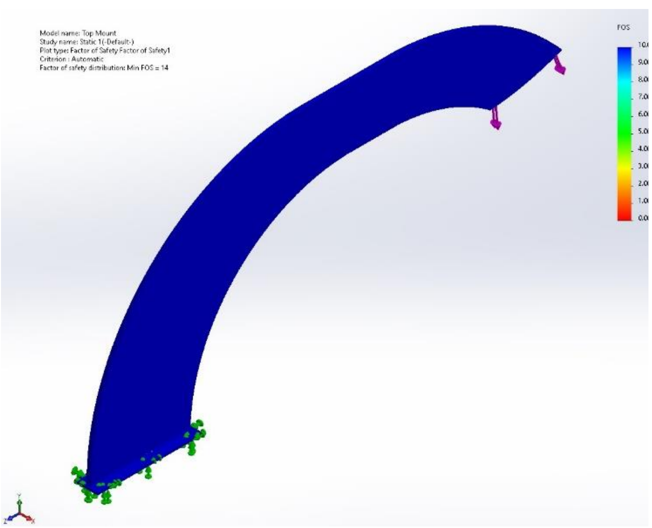

Fig. 19. Factor of Safety Plot Aluminum Alloy 6061

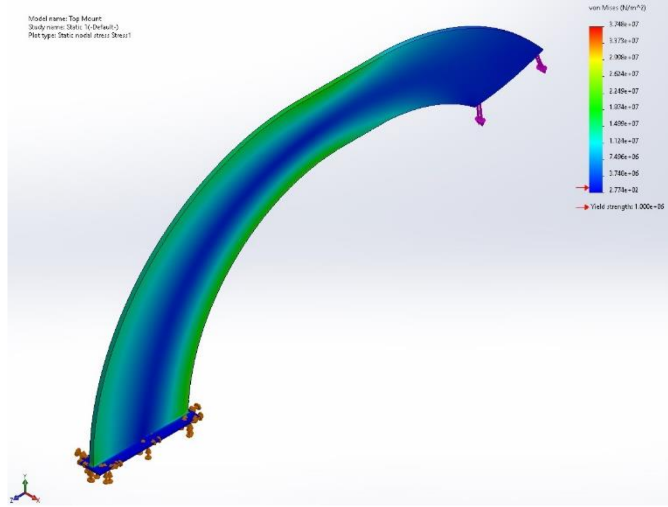

Fig. 20. Stress Plot Carbon Fiber Epoxy

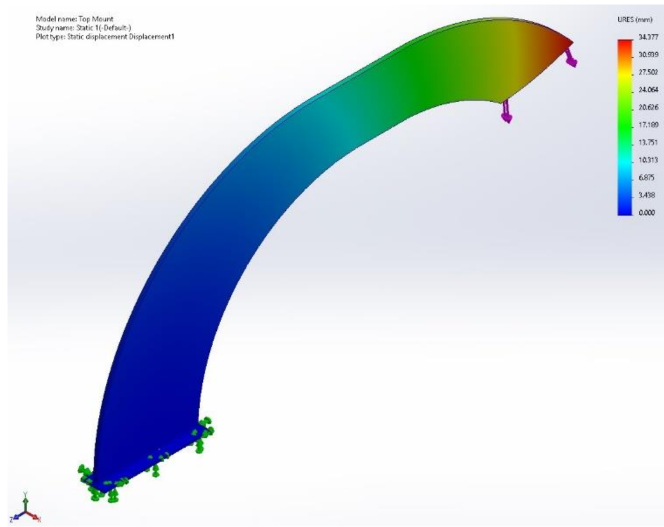

Fig. 21. Displacement Plot Carbon Fiber Epoxy

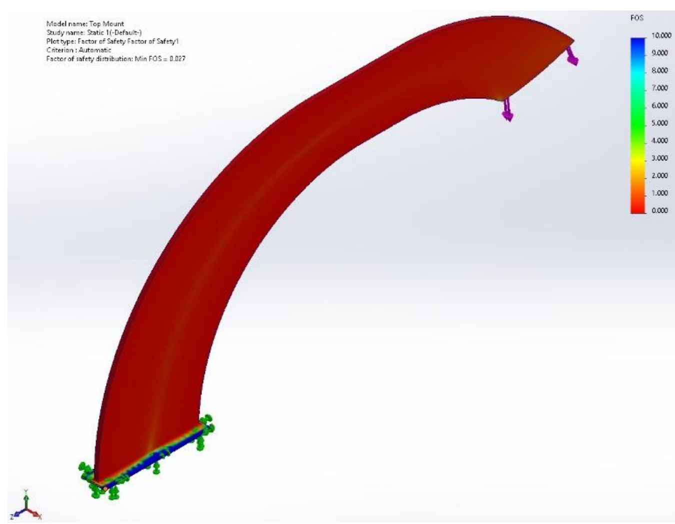

Fig. 22. Factory of Safety Plot Carbon Fiber Epoxy

TABLE II
Results from Aero Loads

Design Style	Stress (GPa)	Displacement (mm)	F.O.S (min)	Weight (grams)
Under Wing – Alloy Steel	0.09881	0.008.442	6.3	1822.22
Under Wing – Aluminum 6061	0.09719	0.02587	0.57	638.96
Under Wing – Carbon Fiber Epoxy	0.1014	0.6442	0.0099	272.15
Swan Neck – Alloy Steel	0.04375	0.4425	14	5407.86
Swan Neck – Aluminum 6061	0.04505	1.346	1.2	1896.26
Swan Neck – Carbon Fiber Epoxy	0.03748	34.377	0.027	807.67

From the data extracted from the study, the two different types of rear spoiler mounts can adequately support the 800 N of force but are only able to support without failure when made of alloy steel. This is shown when we view the minimum factor of safety achieved by the designs. While the swan neck in both alloy steel and 6061 aluminum have factor of safety over 1. With the alloy steel having the best factor of safety in the group, the inherent design of the swan neck in this test displaces more than the standard mount in the same material. The swan neck does have advantages in that it does not experience the same stress the standard mount setup undergoes. With all three material types experiencing on average 50% less stress than their standard mount counterparts. The displacement of the swan neck mounts is on average double that of the displacement of the standard mount. This can be explained by the physics of each mount. While the standard mount has a very small moment arm acting upon it as the wing element sits directly on the mount. The swan neck has a large lever arm inherently in its design as the wing element is mounted further away from the base mounts. The carbon fiber epoxy composite did the worst in these tests, except for the stress of the swan neck, which was the best out of all materials and styles tested.

What can be drawn from these results is that improvements in the design of both stanchion types could see greater improvements for all. The standard under wing design is by far the lightest tested, while unsurprisingly the heaviest is the swan neck, due to the inherent design approach. Improvements in the swan neck design could see the displacement reduced and increase the factor of safety for the aluminum swan neck lead the rest except for weight. Which would then see the swan neck with the lowest stress, lowest displacement, highest factor of safety with only a marginal increase in mass over that of the alloy steel standard mount. This does not include the benefits of the swan neck when comparing the aerodynamic efficiency increase of the wing element due to the mounting position of the swan neck design. The mixture of aluminum and carbon fiber epoxy composite could see improvements far beyond that of either aluminum or carbon fiber epoxy composite alone. Potentially minimizing the weight of the swan neck design and giving the increase in strength the carbon fiber epoxy composite needs to be able to be used in this kind of application.

This study had to have multiple simplifications and assumptions made to be completed. Firstly, the aero load that was tested was derived from a non-standard wing element. As wing designs for rear spoilers are not readily available and would need to be backward engineered to get correct dimensions. Second, the load itself is considered constant. When a vehicle is accelerating the rear spoiler does not experience a constant load, as the load increases and decreases based on the velocity of the vehicle, therefore this load would be a cyclical load. To try and take this into account, the load chosen was that of the average max velocity of a vehicle at the end of a straight on a general racetrack, this being 100mph or about 45 m/s at which the rear spoiler would see its max loading condition. The stanchions are also not subjected to any sideloading and are purely exposed to that of the downward load applied. Thirdly, the design of both stanchions does not consider different vehicles or vehicle types nor are the aero effects of the vehicle considered. The stanchions are generic and have no reference to any real world produced stanchions nor is the wing used for referencing a production wing design.

References

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